

Session 5

Question 1

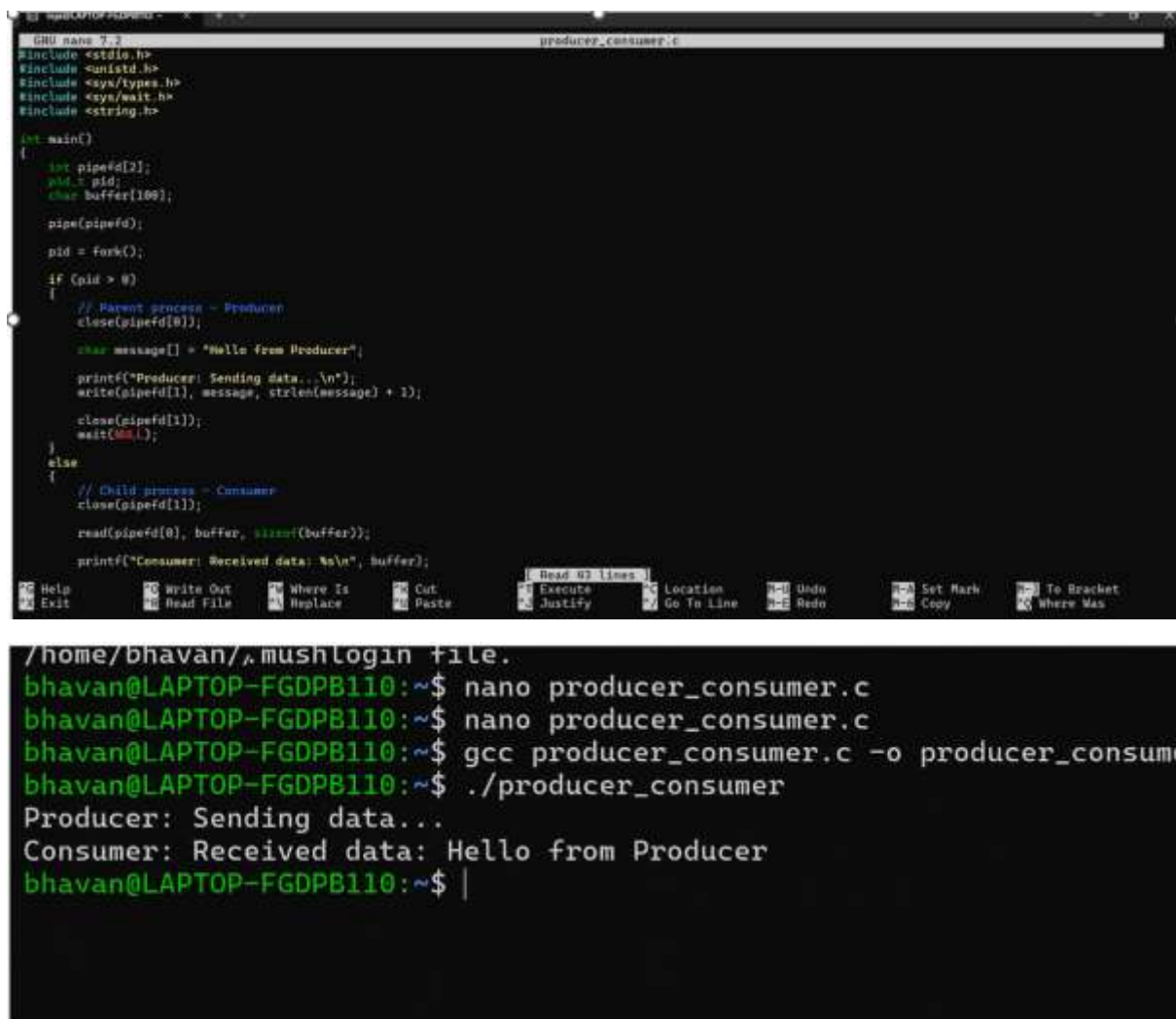
Write a C program to implement the **Producer-Consumer** problem using an anonymous pipe.

Question 2

Write a C program to implement the equivalent of the Linux command:

`ls -l | grep ".c"`

using `fork()`, `pipe()`, `dup2()`, and `execlp()`



```
GNU nano 2.2 producer_consumer.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    int pipefd[2];
    pid_t pid;
    char buffer[100];

    pipe(pipefd);
    pid = fork();
    if (pid > 0)
    {
        // Parent process - Producer
        close(pipefd[0]);

        char message[] = "Hello from Producer";

        printf("Producer: Sending data...\n");
        write(pipefd[1], message, strlen(message) + 1);

        close(pipefd[1]);
        wait(NULL);
    }
    else
    {
        // Child process - Consumer
        close(pipefd[1]);

        read(pipefd[0], buffer, sizeof(buffer));

        printf("Consumer: Received data: %s\n", buffer);
    }
}
```

```
/home/bhavan/.mushlogin file.
bhavan@LAPTOP-FGDPB110:~$ nano producer_consumer.c
bhavan@LAPTOP-FGDPB110:~$ nano producer_consumer.c
bhavan@LAPTOP-FGDPB110:~$ gcc producer_consumer.c -o producer_consumer
bhavan@LAPTOP-FGDPB110:~$ ./producer_consumer
Producer: Sending data...
Consumer: Received data: Hello from Producer
bhavan@LAPTOP-FGDPB110:~$ |
```

Using system calls

```

GNU nano 7.2 shell_equivalent.c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid;

    pipe(pipefd);

    pid = fork();

    if (pid == 0)
    {
        // Child process executes ls -l
        close(pipefd[0]);

        dup2(pipefd[1], STDOUT_FILENO);
        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp");
        exit(1);
    }
    else
    {
        // Parent process executes grep *.c
        close(pipefd[1]);

        dup2(pipefd[0], STDIN_FILENO);
        close(pipefd[0]);

        execlp("grep", "grep", "*.c", NULL);
    }
}

```

Help Exit Write Out Read File Where Is Replace Cut Paste Read 48 lines Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Max

```

Memory usage: 6%          IPv4 address for eth0: 172.18.49.48
Swap usage: 0%

This message is shown once a day. To disable it please create the
/home/bhavan/.hushlogin file.

bhavan@LAPTOP-FGDPB110:~$ nano producer_consumer.c
bhavan@LAPTOP-FGDPB110:~$ nano producer_consumer.c
bhavan@LAPTOP-FGDPB110:~$ gcc producer_consumer.c -o producer_consumer
bhavan@LAPTOP-FGDPB110:~$ ./producer_consumer
Producer: Sending data...
Consumer: Received data: Hello from Producer
bhavan@LAPTOP-FGDPB110:~$ nano shell_equivalent.c
bhavan@LAPTOP-FGDPB110:~$ nano shell_equivalent.c
bhavan@LAPTOP-FGDPB110:~$ gcc shell_equivalent.c -o shell_equivalent
bhavan@LAPTOP-FGDPB110:~$ ./shell_equivalent
bhavan@LAPTOP-FGDPB110:~$ ls -l
-rwxr-xr-x 1 bhavan bhavan 16224 Aug 12 05:54 copy
-rw-r--r-- 1 bhavan bhavan 454 Aug 12 05:52 copy.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 11:37 echo
-rw-r--r-- 1 bhavan bhavan 683 Aug 12 06:38 history.c
-rw-r--r-- 1 bhavan bhavan 1366 Aug 12 06:44 memory.c
-rwxr-xr-x 1 bhavan bhavan 15960 Aug 5 09:15 process
-rw-r--r-- 1 bhavan bhavan 80 Aug 5 09:14 process.c
-rw-r--r-- 1 bhavan bhavan 27 Aug 3 05:49 process.c.save
-rw-r--r-- 1 bhavan bhavan 11 Aug 3 06:00 process.c.save.1
-rwxr-xr-x 1 bhavan bhavan 16360 Aug 26 06:39 producer_consumer
-rw-r--r-- 1 bhavan bhavan 740 Aug 26 06:39 producer_consumer.c
-rw-r--r-- 1 bhavan bhavan 703 Aug 26 06:44 shell_equivalent.c
-rw-r--r-- 1 bhavan bhavan 897 Aug 13 11:25 task1.c
-rw-r--r-- 1 bhavan bhavan 647 Aug 13 11:31 task2.c
-rw-r--r-- 1 bhavan bhavan 683 Aug 13 11:36 task3.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 12:53 task4.c
-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:44 task4.c.save
-rw-r--r-- 1 bhavan bhavan 613 Aug 13 12:54 task4.c.save.1
-rw-r--r-- 1 bhavan bhavan 1179 Aug 13 12:55 task5.c
-rw-r--r-- 1 bhavan bhavan 685 Aug 22 14:05 week3_task3.c
-rw-r--r-- 1 bhavan bhavan 1190 Aug 22 13:32 week4_task1.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:48 week4_task2.c.save
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$

```

```

-rw----- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$ ./shell_equivalent
-rwxr-xr-x 1 bhavan bhavan 16224 Aug 12 05:54 copy
-rw-r--r-- 1 bhavan bhavan 454 Aug 12 05:52 copy.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 11:37 echo
-rw-r--r-- 1 bhavan bhavan 683 Aug 12 06:38 history.c
-rw-r--r-- 1 bhavan bhavan 1366 Aug 12 06:44 memory.c
-rwxr-xr-x 1 bhavan bhavan 15960 Aug 5 09:15 process
-rw-r--r-- 1 bhavan bhavan 80 Aug 5 09:14 process.c
-rw----- 1 bhavan bhavan 27 Aug 3 05:49 process.c.save
-rw----- 1 bhavan bhavan 11 Aug 3 06:00 process.c.save.1
-rwxr-xr-x 1 bhavan bhavan 16360 Aug 26 06:39 producer_consumer
-rw-r--r-- 1 bhavan bhavan 740 Aug 26 06:39 producer_consumer.c
-rw-r--r-- 1 bhavan bhavan 703 Aug 26 06:44 shell_equivalent.c
-rw-r--r-- 1 bhavan bhavan 897 Aug 13 11:25 task1.c
-rw-r--r-- 1 bhavan bhavan 647 Aug 13 11:31 task2.c
-rw-r--r-- 1 bhavan bhavan 683 Aug 13 11:36 task3.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 12:53 task4.c
-rw----- 1 bhavan bhavan 599 Aug 13 11:44 task4.c.save
-rw-r--r-- 1 bhavan bhavan 613 Aug 13 12:54 task4.c.save.1
-rw-r--r-- 1 bhavan bhavan 1179 Aug 13 12:55 task5.c
-rw-r--r-- 1 bhavan bhavan 685 Aug 22 14:05 week3_task3.c
-rw-r--r-- 1 bhavan bhavan 1190 Aug 22 13:32 week4_task1.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c
-rw----- 1 bhavan bhavan 607 Aug 22 13:48 week4_task2.c.save
-rw----- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$ |

```

```

bhavan@LAPTOP-FGDPB110:~$ ls
OSLab  copy.c  memory  process.c.save  shell_equivalent  task2  task4  task4.save  week3_task3.c  week4_task2.c
OSSP  echo  memory.c  process.c.save.1  shell_equivalent.c  task2.c  task4.c  task5  week4_task1  week4_task2.c.save
week1  history.c  process  producer_consumer  task1  task3  task4.c.save  task5.c  week4_task1.c  week4_task2.c.save.1
copy  ls  process.c  producer_consumer.c  task1.c  task3.c  task4.c.save.1  week3_task3  week4_task2  week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$ ls -l
total 296
drwxr-xr-x 2 bhavan bhavan 4096 Aug 8 06:15 OSLab
drwxr-xr-x 11 bhavan bhavan 4096 Aug 7 06:00 OSSP
drwxr-xr-x 3 bhavan bhavan 4096 Aug 7 05:50 Week3
-rwxr-xr-x 1 bhavan bhavan 16224 Aug 12 05:54 copy
-rw-r--r-- 1 bhavan bhavan 454 Aug 12 05:52 copy.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 11:37 echo
-rw-r--r-- 1 bhavan bhavan 683 Aug 12 06:38 history.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 11:25 ls
-rwxr-xr-x 1 bhavan bhavan 15960 Aug 12 06:45 memory
-rw-r--r-- 1 bhavan bhavan 1366 Aug 12 06:44 memory.c
-rwxr-xr-x 1 bhavan bhavan 15960 Aug 5 09:15 process
-rw-r--r-- 1 bhavan bhavan 80 Aug 5 09:14 process.c
-rw-r--r-- 1 bhavan bhavan 27 Aug 3 05:49 process.c.save
-rw----- 1 bhavan bhavan 11 Aug 3 06:00 process.c.save.1
-rwxr-xr-x 1 bhavan bhavan 16360 Aug 26 06:39 producer_consumer
-rw-r--r-- 1 bhavan bhavan 740 Aug 26 06:39 producer_consumer.c
-rwxr-xr-x 1 bhavan bhavan 16224 Aug 26 06:44 shell_equivalent
-rw-r--r-- 1 bhavan bhavan 703 Aug 26 06:44 shell_equivalent.c
-rwxr-xr-x 1 bhavan bhavan 16184 Aug 13 11:25 task1
-rw-r--r-- 1 bhavan bhavan 897 Aug 13 11:25 task1.c
-rwxr-xr-x 1 bhavan bhavan 16136 Aug 13 11:31 task2
-rw-r--r-- 1 bhavan bhavan 647 Aug 13 11:31 task2.c
-rwxr-xr-x 1 bhavan bhavan 16104 Aug 13 11:36 task3
-rw-r--r-- 1 bhavan bhavan 683 Aug 13 11:36 task3.c
-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:50 task4
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 12:53 task4.c
-rw----- 1 bhavan bhavan 599 Aug 13 11:44 task4.c.save
-rw-r--r-- 1 bhavan bhavan 613 Aug 13 12:54 task4.c.save.1
-rwxr-xr-x 1 bhavan bhavan 16080 Aug 13 12:55 task5
-rw-r--r-- 1 bhavan bhavan 1179 Aug 13 12:55 task5.c
bhavan@LAPTOP-FGDPB110:~$ |

```

```

-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:50 task4
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 12:53 task4.c
-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:44 task4.c.save
-rw-r--r-- 1 bhavan bhavan 613 Aug 13 12:54 task4.c.save.1
-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:49 task4.save
-rwxr-xr-x 1 bhavan bhavan 16480 Aug 13 12:55 task5
-rw-r--r-- 1 bhavan bhavan 1179 Aug 13 12:55 task5.c
-rwxr-xr-x 1 bhavan bhavan 16176 Aug 22 14:06 week3_task3
-rw-r--r-- 1 bhavan bhavan 685 Aug 22 14:05 week3_task3.c
-rwxr-xr-x 1 bhavan bhavan 16360 Aug 22 13:35 week4_task1
-rw-r--r-- 1 bhavan bhavan 1190 Aug 22 13:32 week4_task1.c
-rwxr-xr-x 1 bhavan bhavan 16216 Aug 22 13:51 week4_task2
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:48 week4_task2.c.save
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$ ls -l | grep ".c"
-rw-r--r-- 1 bhavan bhavan 16224 Aug 12 05:54 copy
-rw-r--r-- 1 bhavan bhavan 454 Aug 12 05:52 copy.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 11:37 echo
-rw-r--r-- 1 bhavan bhavan 683 Aug 12 06:38 history.c
-rw-r--r-- 1 bhavan bhavan 1366 Aug 12 06:44 memory.c
-rwxr-xr-x 1 bhavan bhavan 15960 Aug 5 09:15 process
-rw-r--r-- 1 bhavan bhavan 80 Aug 5 09:14 process.c
-rw-r--r-- 1 bhavan bhavan 27 Aug 3 05:49 process.c.save
-rw-r--r-- 1 bhavan bhavan 11 Aug 3 06:00 process.c.save.1
-rwxr-xr-x 1 bhavan bhavan 16360 Aug 26 06:39 producer_consumer
-rw-r--r-- 1 bhavan bhavan 740 Aug 26 06:39 producer_consumer.c
-rw-r--r-- 1 bhavan bhavan 703 Aug 26 06:44 shell_equivalent.c
-rw-r--r-- 1 bhavan bhavan 897 Aug 13 11:25 task1.c
-rw-r--r-- 1 bhavan bhavan 647 Aug 13 11:31 task2.c
-rw-r--r-- 1 bhavan bhavan 683 Aug 13 11:36 task3.c
-rw-r--r-- 1 bhavan bhavan 0 Aug 13 12:53 task4.c
-rw-r--r-- 1 bhavan bhavan 599 Aug 13 11:44 task4.c.save
-rw-r--r-- 1 bhavan bhavan 613 Aug 13 12:54 task4.c.save.1
-rw-r--r-- 1 bhavan bhavan 1179 Aug 13 12:55 task5.c
-rw-r--r-- 1 bhavan bhavan 685 Aug 22 14:05 week4_task3.c
-rw-r--r-- 1 bhavan bhavan 1190 Aug 22 13:32 week4_task1.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:48 week4_task2.c.save
-rw-r--r-- 1 bhavan bhavan 607 Aug 22 13:50 week4_task2.c.save.1
bhavan@LAPTOP-FGDPB110:~$ |

```

Theory

A **pipe** is an Inter-Process Communication (IPC) mechanism used to exchange data between processes. A pipe has two ends: one for reading and one for writing.

The `fork()` system call creates a new child process. In the Producer-Consumer program, the parent process acts as the producer and writes data into the pipe, while the child process acts as the consumer and reads the data from the pipe.

The `pipe()` system call creates the communication channel. The `write()` system call sends data through the pipe, and the `read()` system call receives the data.

The `dup2()` system call is used to redirect standard input or output to the pipe. The `execlp()` system call is used to execute commands such as `ls` and `grep`.

The command `ls -l | grep ".c"` displays detailed information about files whose names contain `.c`. The output of `ls -l` is passed through the pipe to `grep`.

Thus, pipes provide a simple and effective way for processes to communicate and exchange data.